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Evaluating socioeconomic and equity outcomes in livelihood interventions: Insights from Community Resource Management Areas (CREMAs) in northern Ghana

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ABSTRACT

Over the past 25 years, Ghana's Community Resource Management Area (CREMA) program has expanded to cover 2 million hectares across 600 communities, aiming to achieve its dual objectives of biodiversity conservation and poverty reduction. As Ghana's flagship Community-Based Natural Resource Management (CBNRM) initiative, it has attracted significant investment from international organizations for livelihood interventions to reduce poverty and incentivize conservation. Yet little is known about how these interventions are implemented or who benefits. Using qualitative methods, this study examines stakeholders' perceptions of the types, socioeconomic impacts, and equity dynamics of livelihood interventions in three CREMAs in northern Ghana. Our findings reveal that interventions that enhance existing livelihoods, such as modern beekeeping, remain the most frequently implemented. Although the socioeconomic impacts of these interventions vary across and within CREMAs, they provide individuals with modest, seasonal income and support community development projects. The gender dimensions of the interventions show that gender-targeted, pro-poor distribution norms have increased women's inclusion and access to cash income but also raise concerns about exclusion and intersectional inequities. By situating our findings within debates on distributional and contextual equity, we offer insights into how pro-poor norms interact with gender and other contextual factors to shape economic and equity outcomes in incentive-based conservation. This article is relevant beyond the case study, as it can inform the design of equitable, context-sensitive, and sustainable livelihood interventions in current and future incentive-based conservation efforts across the tropics and beyond.

1. Introduction

The long history of the deprivation of use rights associated with protected area establishment led to resentment of biodiversity conservation among local communities (Garland, 2008; Vasquez and Sunderland, 2023). Access and use rights to forests and their resources enjoyed by local communities prior to the imposition of fortress conservation systems were either completely restricted or significantly reduced (Duffy, 2010). The mid-1980s saw increased recognition of local communities as owners and guardians of customary territories and greater efforts to rectify decades of displacement and exclusion from these territories (Dressler et al., 2010). Community-based natural resource management (CBNRM) emerged as a conservation approach that empowers communities to collaboratively manage their natural resources to protect biodiversity and spur livelihoods and economic growth in rural areas (Adeyanju et al., 2021; Foli et al., 2018; Murphree, 2009). Variations of CBNRM have been rolled out in almost all regions of the world, including Africa, Asia, Latin America, and North America (Dressler et al., 2010; Bulkan et al., 2022). Some of the earliest CBNRM programs in Africa were established in southern Africa, including

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Administrative Management Design for Game Management Areas (ADMADE) in Zambia, the Communal Areas Management Program for Indigenous Resources (CAMPFIRE) in Zimbabwe, and communal conservancies in Namibia (Child and Barnes, 2010).

Ghana's official CBNRM program began in 2000 with the introduction of the Community Resource Management Areas (CREMAs) program by the Wildlife Division of the Ghana Forestry Commission. CREMAs were established to encourage active local participation in natural resource management, leading to biodiversity conservation and improved livelihoods (Adeyanju et al., 2021; Asare et al., 2013; Forestry Commission of Ghana (FC), 2004). CREMAs are often areas with significant forest and wildlife resources, where communities collaborate to manage their shared natural resources sustainably (Agyare, 2017). CREMAs are promoted as a successful model of community-based natural resource management in Ghana and beyond (Adeyanju et al., 2021; Agyare, 2017), with at least 59 CREMAs registered at the Wildlife Division of the Ghana Forestry Commission (World Bank, 2024). A total of 45 CREMAs have been mapped, and many others are at various stages of development (World Bank, 2024).

As with many CBNRM programs, the CREMA scheme largely depends on donors and non-governmental organizations (NGOs) for initial and ongoing funding and technical support (Adeyanju et al., 2021; Baruah et al., 2016; see Appendix A in the Supplementary material for key ongoing or completed donor-funded projects in Ghanaian CREMAs). These external actors (donors and NGOs), including the United Nations Development Program (UNDP), International Union for Conservation of Nature (IUCN), and Arocha Ghana, play an important role in financing livelihood interventions (e.g., ecotourism, shea nut certification, and beekeeping) to reduce poverty and unsustainable resource exploitation (Agyare et al., 2015a; Baruah et al., 2016; Dzekoto and Bosu, 2018; Gilli et al., 2020). Hence, most communities anticipate improved livelihoods from participating in the CREMA program (Agyare et al., 2015a; Bempah et al., 2019; Gilli et al., 2020).

Prior research has examined the conservation, economic, and sociocultural outcomes of the CREMA program (Agyare et al., 2015a, 2015b, 2024). However, these studies have primarily relied on structured survey data, which, while valuable for generalization, are often inadequate to produce rich, contextual data on complex topics (Newing et al., 2011). Rural livelihoods are complex, diverse, and dynamic, shaped by intersecting political and environmental processes across scales (Scoones, 2009). Moreover, existing scholarship tends to treat livelihood interventions – the main mechanisms for achieving CREMA's economic objectives – as secondary to conservation outcomes, rather than the focus of analysis (Bayala et al., 2024; Bempah et al., 2019). Few studies have critically evaluated these interventions, and existing evidence suggests that CREMA communities have achieved only limited economic empowerment (Mawutor and Hajjar, 2024). This lack of focused analysis represents a critical gap in understanding how CREMAs deliver on their livelihood goals. Our study addresses these shortcomings by characterizing the nature, scope, impacts, and dynamics of livelihood interventions within three CREMAs in northern Ghana. Specifically, we ask: (1) how can various livelihood interventions implemented within CREMAs be categorized; (2) how do key stakeholders—community members, government officials, NGOs, and donors—perceive the socioeconomic impacts of livelihood interventions within CREMAs; and (3) how are the benefits of livelihood interventions distributed within and across CREMAs, by gender, and what factors determine the extent of the impact of livelihood interventions? By addressing these questions, the study provides the empirical depth policymakers and practitioners need to understand how introduced interventions have affected participating communities. Ultimately, this work informs evidence-based policies to guide the design and effective integration of livelihood interventions and poverty-reduction strategies into CREMAs and similar programs across the region.

We present the conceptual background in Section 2. Section 3 outlines the methodology, including the study context (Section 3.1), case study selection (Section 3.2), and data collection and analysis (Section 3.3). Sections 4 and 5 present and synthesize the results and discuss the findings. Section 6 concludes the article and offers recommendations for policy and practice.

2. Conceptual framework

This paper draws on the literature on incentive-based conservation (Charnley, 2023; Suich, 2013; Wright et al., 2016). Past conservation program failures have often been attributed to a lack of tangible benefits for local communities (Chambers et al., 2022; Duffy, 2010). In response, conservation organizations have shifted toward incentive-based mechanisms, such as alternative livelihood programs and market-based approaches, to address human well-being and other development goals alongside conservation objectives (Hajjar et al., 2021; Roe et al., 2015; Wicander and Coad, 2018).

Most CREMAs in Ghana operate as incentive-based conservation initiatives (World Bank, 2024; see Section 3.2) and meet the four criteria outlined by Charnley (2023): (1) they are formally established through an external initiative; (2) the initiative is driven by forest and wildlife conservation goals; (3) they alter current or customary forest access, use, and management to facilitate conservation; and (4) extrinsic incentives to participate are part of the initiative. At the core of these initiatives are livelihood interventions designed to generate economic benefits and meet local needs (Roe et al., 2015; Wright et al., 2016).

The impact of incentive-based conservation approaches on local livelihoods varies widely and can be positive, negative, or neutral (Charnley, 2023; Roe et al., 2015; Suich, 2010). Even when outcomes are positive, benefits often accrue at the community or higher levels rather than reaching poor and marginalized households (Hajjar et al., 2021; McDermott and Schreckenberg, 2009). These disparities have brought social equity considerations to the forefront of incentive-based conservation approaches (Friedman et al., 2022; Klein et al., 2015; Law et al., 2018; McDermott and Schreckenberg, 2009).

Equity in this context is multidimensional, encompassing procedural, distributional, recognitional, and contextual elements (Law et al., 2018). Without equitable participation (procedural equity), marginalized groups (i.e., across gender, age, social status, ethnicity, and wealth) risk exclusion, enabling elite capture of benefits (distributional equity) (Friedman et al., 2022; Leisher et al., 2016). By respecting local norms, knowledge systems, and rights, incentive-based conservation approaches can better respond to the varied aspirations and needs of local people (recognitional equity) (Sene-Harper et al., 2019; Wright et al., 2016). Finally, the social, political, economic, and cultural contexts in which these approaches are implemented critically influence their outcomes (contextual equity) (Hajjar et al., 2021; Klein et al., 2015; Law et al., 2018). Ignoring these contextual factors can undermine both conservation and livelihood objectives.

Our work specifically analyses the distributional equity dimension of livelihood interventions within CREMAs, examining how benefits are apportioned across genders and what factors shape these outcomes. By doing so, it contributes to the literature on incentive-based conservation by linking equity and contextual considerations to the effectiveness of livelihood interventions.

3. Methodology

3.1. Study context

Ghana is a West African country rich in natural resources, including gold, bauxite, and economically valuable tree species such as shea (*Vitellaria paradoxa*), dawadawa (*Parkia biglobosa*) and many other species which are often incorporated into agroforestry production systems. Cocoa and, to a lesser extent, coffee are widely cultivated, notably in the country's humid region. One-third of Ghana's landmass is in northern Ghana, the region most vulnerable to climate change because

of low socioeconomic development and reliance on rainfed agriculture (Antwi-Agyei et al., 2012; UNDP, 2018). Despite lower population density than in southern Ghana, the incidence and intensity of multidimensional poverty in northern Ghana are high. For instance, the Northern Region had the highest proportion of multidimensionally poor people at 80.8%, followed by the Upper East Region at 68%, and the Upper West Region at 65.5%. In contrast, the Greater Accra and Ashanti Regions in southern Ghana registered the lowest levels of poverty (22.5% and 31.1%, respectively) (Ghana Statistical Service, 2020). This high poverty rate is attributed to historical neglect by colonial authorities, stemming from a perceived absence of valuable resources, a factor that has also shaped limited investment by post-independence governments in the region (Nyaaba and Bob-Milliar, 2019). The widespread poverty has led to the rise and clustering of local and international NGOs engaging in development interventions in northern Ghana (Adjei et al., 2012; Bob-Milliar and Alagidede, 2022), including those establishing CREMAs and alternative livelihood projects (Baruah, 2015; Baruah et al., 2016). Despite these efforts, few development interventions have yielded positive results in alleviating regional poverty (Nyaaba and Bob-Milliar, 2019; UNDP, 2018).

The multi-ethnic, multilingual, and multi-religious peoples of northern Ghana live in the five northernmost administrative regions: the Upper East, Upper West, Savannah, Northeast, and Northern Regions (Bob-Milliar and Alagidede, 2022). Most depend on agriculture and natural resources for their livelihoods, which puts significant pressure on the broader ecosystem (Marchetta, 2011; UNDP, 2018). Community members gather a wide variety of non-timber forest products (NTFPs) for subsistence and sale at periodic local markets (Wardell and Fold, 2013). The most common NTFPs in this region include shea nuts, wood fuels, thatching materials, grasses for basketry, vegetable dyes, building poles, medicinal plants, and bush meat. Specifically, shea nuts and shea butter play a crucial role in sustaining rural livelihoods, particularly for women who supply the burgeoning domestic and international markets (Gilli et al., 2020; Wardell and Fold, 2013).

3.2. Case study selection

To address our research questions on the scope, impacts, and dynamics of livelihood interventions within CREMAs, we purposively selected three CREMAs in northern Ghana (Fig. 1). The selected CREMAs share attributes generally associated with CREMAs in Ghana. First, most CREMAs have been established through financial and technical support from donor-funded projects or organizations (Adeyanju et al., 2021; Baruah et al., 2016). All three sites were created by or received significant financial and technical support from donor-funded projects or organizations. The Sanyiga Kasena Gavara Kara (SKGK) CREMA was created through funding from the World Bank's Sustainable Land and Water Management Project (SLWMP) (World Bank, 2021). Zukpiri CREMA was established with support from the Global Environment Facility (GEF), the United Nations Development Program's Small Grant Program (UNDP SGP), and the Nadowli District Assembly (Baddianaah and Baaweh, 2021). Murugu Mognori was established with funds from the Dutch government under the Wildlife Division Support Project (2002–2007) (World Bank, 2024). Second, most CREMAs are located near vital natural resource areas and were created to protect aquatic and terrestrial flora and fauna (Asare et al., 2013). SKGK is located within the Western Wildlife Corridor, created to facilitate wildlife movement across Ghana and Burkina Faso (Bayala et al., 2020); Zukpiri is located near the Black Volta River, which forms a boundary between Ghana and Burkina Faso and holds significant ecological, economic, and social importance (Baddianaah and Baaweh, 2021); Murugu Mognori is one of the fringe communities of Mole National Park (Ghana's largest and oldest national park) (Dzekoto and Bosu, 2018). Third, most CREMAs have an active presence of one or more NGOs or community-based organizations. SKGK has the Organization for Indigenous Initiatives and Sustainability (ORGIIS Ghana) (Bayala et al., 2020); Zukpiri has Zintang

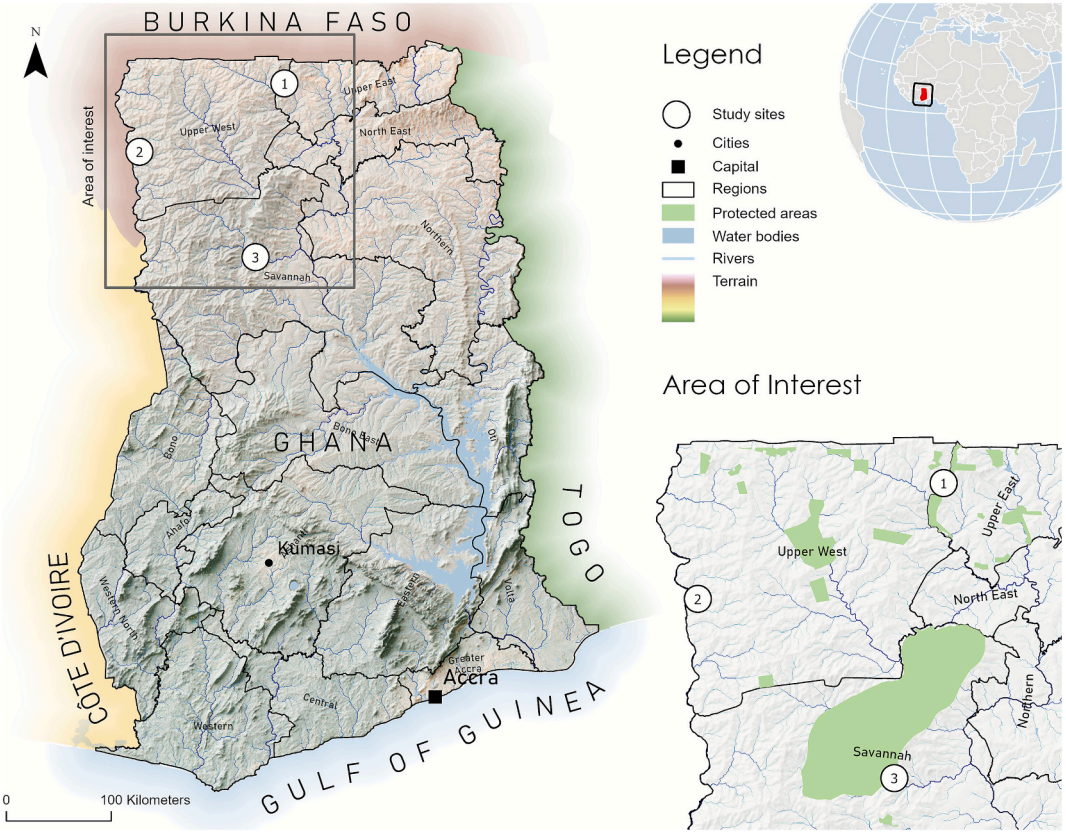


Fig. 1. Map of Ghana showing the study sites. *Source:* Created by Jeremy Allen, University of Columbia, Canada.

Traditional Healers Association (Baddianaah and Baaweh, 2021); and Murugu Mognori has A Rocha Ghana (Dzekoto and Bosu, 2018).

To ensure the representativeness of the CREMAs, we selected cases that varied in their year of establishment and number of constituent communities (see Table 1). All selected CREMAs were active, allowing us to approach stakeholders to share their experiences. The Sanyiga Kasena Gavara Kara (SKGK) CREMA covering the Upper East and Upper West Regions, the Zukpiri CREMA in the Upper West Region, and the Murugu–Mognori CREMA in the Savannah Region (Table 1; Fig. 1). We selected two communities from each of the three CREMAs ($N = 6$) based on recommendations from local stakeholders and ease of access to the CREMA communities.

3.3. Data collection and analysis

The first author made two scoping trips to northern Ghana in April and September 2022 to visit the research communities and participated in two multistakeholder workshops on integrated landscape management in the region, in which the co-authors also participated (see Reed et al., 2024). The first author met with research partners and local stakeholders, gained insight into proposed study sites, and pilot-tested the interview questions.

Data collection was conducted in July and August 2024. We used qualitative methods, including direct observation, interviews, focus group discussions (FGDs), and secondary data sources (government, NGO and donor reports, CREMA reports and websites) (Creswell and Poth, 2025). Perceptions of conservation policy and practice were used to gain insights into the dynamics and impacts of the CREMA program and its associated livelihood interventions (Bennett, 2016). Hence, our qualitative data collection focused on examining stakeholders' perceptions rather than collecting objective quantitative evidence (Bennett, 2016). Although quantitative methods and data are often considered superior to qualitative data in conservation writings (Adams and Sandbrook, 2013), we affirm that qualitative methods are 'a valuable and valid approach to matters critical to conservation research,' especially for understanding complex social concepts such as perceptions and attitudes (Drury et al., 2011, p.18).

We conducted 80 interviews and 12 focus group discussions (Table 2), using purposive and snowball sampling (Newing et al., 2011). Purposive sampling ensured that community members with knowledge of the CREMA program were selected from diverse resource user groups (farmers, hunters, shea and baobab collectors and processors) and local leaders (CREMA executives, local chiefs, earth priests, and assembly-persons) across the six communities. We also identified other participants with relevant knowledge and perspectives on the topic. To mitigate potential sampling biases, interviews and focus groups were triangulated with key informant interviews and firsthand observations, and strategies such as open-ended questions, an interview funnel, and probing were used to minimize response bias (Newing et al., 2011).

Across the three selected CREMAs, 61 semi-structured interviews and 12 focus groups were held in the six communities, with men and women divided into separate focus groups in each community (participation ranged from 5 to 8 people). The interviews and focus groups were conducted primarily in the local language of each CREMA (*Kasem* in SKGK CREMA, *Dagaari* in Zukpiri CREMA, and *Hanga* in Murugu-Mognori CREMA) and were translated into English on-site with the help of three research assistants. We also conducted in-depth semi-structured interviews with 19 key informants with experience in CREMA implementation, including government agencies ($n = 10$), NGOs and donor organizations ($n = 8$), and a private commodity company (n = 1; see Appendix B in the Supplementary material for the roles and organizations of the institutional actors interviewed). Interviews lasted between 15 minutes and 2 hours, based on the participant's knowledge of the CREMA programs and activities. The interviews addressed topics such as the CREMA establishment process and ongoing management, as well as the types, implementation processes, and impacts of livelihood interventions (see Appendix C in the Supplementary material for the

Table 1
Characteristics of the selected CREMAs.

	Sanyiga Kasena Gavara Kara (SKGK) CREMA	Zukpiri CREMA	Murugu–Mognori CREMA
Administrative region(s)	Upper East and Upper West	Upper West	Savannah
CREMA significance	One of the six official CREMAs in the Western Wildlife Corridor, which facilitates/ enables the migration of wild animals from Burkina Faso to Ghana.	One of the few CREMAs established by a community-based organization (CBO) and rated first out of the 18 established CREMAs in Northern Ghana.	Among the earliest CREMAs established in Ghana and the first in the northern region surrounding Mole National Park (Ghana's oldest and largest national park).
Constituent communities	10	16	2
Year established/ Certificate of devolution	2016/2019	2009/2011	2008
Size	5870 ha	4200 ha	23,377 ha
Establishment facilitated by (or current NGO partner organization)	The Wildlife Division and The Organization for Indigenous Initiatives and Sustainability (ORGIIS Ghana)	Zintang Traditional Healers Association	The Wildlife Division and A Rocha Ghana
Motivation for CREMA	Sustainably manage natural resources through green value chains such as the shea and baobab trade.	Conserve and improve traditional medicine sources.	Formalize use and access to shea land and then benefit from the added economic value of certified organic shea trade.

Sources: Compiled by the authors based on published literature (Agyare et al., 2015a; Asare et al., 2013; Baddianaah and Baaweh, 2021; Bayala et al., 2020; Dzekoto and Bosu, 2018).

Table 2
Overview of research participants.

S/N	Category	Interviews	Number of FGDs
1.	SKGK CREMA	Men: 13 Women: 6	4
	• Nakong • Basisan		
2.	Zukpiri CREMA	Men: 11 Women: 11	4
	• Siiru • Meguo		
3.	Murugu-Mognori CREMA	Men: 10 Women: 10	4
	• Murugu • Mognori		
	Total for CREMAs	Men: 34 Women: 27	12
4.	Municipal government	1	
5.	District government	2	
6.	National/Regional government agency	7	
	Total for Government	10	
7.	Non-Governmental Organizations (NGOs)	7	
8.	Donor organization	1	
	Total for NGOs and donor organization	8	
9.	Private commodity company	1	
	Total for private sector	1	
	Grand total	80	12

interview and focus group guides). In addition, the first author engaged in participant observation during tourist visits, shea butter production processes, the distribution of donor-funded items, and daily life within the communities.

Data collection continued until saturation was reached (i.e., no new insights were provided by participants, or no new meaning was drawn from the data) (Braun and Clarke, 2021). Participants provided consent for audio recording of the interviews and focus group discussions. We obtained informed consent and complied with the ethical principles of research, including guaranteeing safety and anonymity. The audio-recorded data were transcribed into text and then subjected to thematic analysis in NVivo Qualitative Data Analysis software. We applied simultaneous deductive and inductive (hybrid) coding to integrate pre-defined codes with emergent ones (Fereday and Muir-Cochrane, 2006). Participant quotes were used to support the themes in the results section.

4. Results

Here, we discuss the types of livelihood interventions, their impacts, and other key findings related to gender, equity, benefit sharing, and the reach of impact.

4.1. Livelihood interventions in CREMAs

We identified 21 distinct livelihood interventions in the six CREMA case study communities (Table 3; see Appendix D in the Supplementary material for a photo collage of livelihood interventions across the CREMAs). Some were no longer in operation due to a lack of capital and damaged machinery (e.g., boreholes, milling machines, tricycles). We categorized the interventions using established typologies in the conservation and value chain literature.[1] We used a typology developed by STAP (2024) to distinguish between alternative, enhanced, and supplemental/diversified livelihood interventions. The interventions identified in the studied CREMAs fall under alternative livelihood interventions (e.g., solar engineering training), enhanced livelihood interventions (e.g., dry season gardening, soap making, sale of organic shea nuts and butter, Village Savings and Loans Association (VSLA)), and supplemental/diversified livelihood interventions (e.g., ecotourism, beekeeping, soap making). These interventions, often donor-funded, vary across communities in implementation and efficacy. However, the six most frequently implemented were shea and other economic tree-related interventions (in terms of enhancing market access, organic certification, and value addition), beekeeping, dry season gardening, tree planting, soap-making training, and provision of mixed transport options for conveying people and goods (e.g., bicycles, motorcycles, tricycles, canoes, donkeys, and donkey carts). Overall, beekeeping and tree-planting programs are the two most widely implemented interventions across all six communities.

Traditionally, honey production in this region involves men harvesting honey from wild bee colonies by felling trees and using fire to kill the bees, which frequently leads to bushfires, deforestation, and the destruction of the bee colonies. The modern beekeeping method, introduced by conservation and development NGOs, uses beehives, which prevent bushfires and tree felling while also providing income for beekeepers through the production of honey and beeswax.

A significant number of livelihood interventions in CREMAs have also focused on conservation, restoration, market access, and value addition for shea and other economic trees. Given the economic importance of these trees to women's livelihoods, project implementers have made significant efforts to increase awareness of the economic and conservation value of shea and other economic trees. Many participants reported that CREMA laws and patrols have helped increase resource protection and law enforcement, thereby reducing tree cutting to some extent. Tree nurseries have been established and are managed by local people, with donor funding, to produce seedlings for transplanting across the region. CREMA members are regularly supplied with economic tree species such as cashew (*Anacardium occidentale*), guava (*Psidium guajava*), locust bean (*Parkia biglobosa),* baobab (*Adansonia digitata*), and timber species such as *Acacia* spp., to plant on farms and within the community. Some interventions have provided market access for shea nuts and other economic trees, such as baobab, locust bean, and velvet tamarind (*Dialium guineense*), through guaranteed wholesale purchases at the community level.

In the finance category, conservation NGOs have adopted the VSLA model to address poverty and improve livelihoods. The VSLA—locally known as a *susu* group—is a community-led microfinance initiative widely recognized in rural Ghana for enhancing savings and providing loans and other financial services to its members (Pienaah and Luginaah, 2024). It operates as a simple yet well-structured system, bringing together groups of 10–30 people who meet weekly to save and borrow money at low interest rates. Over a 52-week cycle, savings and profits are distributed among the members (Pienaah and Luginaah, 2024). For example, ORGIIS Ghana—the Organization for Indigenous Initiatives and Sustainability, an NGO actively working with the SKGK CREMA—uses the VSLA as an entry point to engage with communities, followed by value-chain development for commodities such as shea and baobab. Across the 10 communities that make up SKGK CREMA, there are 56 susu groups, each averaging 27 members.

Several skill development training programs on existing and new skills, such as soap making and solar engineering, have been implemented across the CREMAs. However, these initiatives often stop at knowledge transfer, as only a few training programs provide participants with start-up funding to translate their skills into commercial ventures. For example, in the Meguo and Siiru communities, although numerous women have been trained in soap making, only about three women in each community currently produce soap for commercial purposes because of insufficient funds to purchase the necessary equipment and soap-making ingredients. In the Zukpiri CREMA, three women without formal education were sent to India for training in solar power repair and engineering.

Project implementers, particularly NGOs working with the CREMAs, often prefer certain types of interventions over others. An NGO staff member shared the rationale guiding their decisions:

> "The knowledge and skills have been passed down through our ancestors, and the women themselves were already engaging in one or two of these value chains at a smaller scale. What we are doing is enhancing the scale. It is already an existing value chain with weak linkages. We are improving their capacity to access organic markets and to negotiate and secure acceptable prices." [Donor /NGO staff interviewee 4].[2]

It is evident that, for most NGOs, the underlying rationale is to pursue conservation goals alongside or through rural economic development objectives.

4.2. Socio-economic impacts of interventions

In CREMAs, the socioeconomic impact of livelihood interventions spans multiple dimensions at both the individual and community levels.

[1] Some of the interventions cut across multiple categories in the typology, hence the overlap. For instance, soap making is categorized under supplemental/diversified as well as enhanced livelihood interventions. This is because female respondents mentioned their parents have been making soap using shea butter for generations. So, the interventions came to enhance aspects such as packaging, etc. while also diversifying their income streams.

[2] All quotations were lightly edited for clarity and readability, while preserving the original intent.

Table 3
The diversity of livelihood interventions in the six communities selected from the three CREMAs. Amoah and Mensah, 2023

S/N	Livelihood intervention	CREMAs					
		SKGK		Zukpiri		Murugu-Mognori	
		Nakong	Basisan	Siiru	Meguo	Murugu	Mognori
1.	Apiculture (modern beekeeping)	Both	Both	Both	Both	Both	Both
2.	Agriculture development & support		Both	Both	Both	Men	Both
3.	Establishment of dry-season gardens		Women	Women	Women		Both
4.	Poultry birds and small ruminant rearing			Women	Women	Both	
5.	Ecotourism						Both
6.	Tree planting	Both	Both	Both	Both	Both	Both
7.	Shea value addition & trade	Women		Women	Women	Women	
8.	Trade in baobab and other economic fruit trees	Women					
9.	Organic certification			Both	Both		
10.	Organic cassava farming and trade					Both	
11.	Building of a warehouse	Women					Women
12.	Training on pomade making			Women	Women		
13.	Training on soap making			Women	Women		
14.	Provision of transportation		Both	Both	Both	Both	Both
15.	Provision of a grinding mill	Women				Women	Women
16.	Training on the standardization of herbal preparation and value addition			Men	Men		
17.	Oil extraction from seeds			Women	Women		
18.	Batik tie dye			Women	Women		
19.	Establishment of a *susu* group (cooperatives)	Both	Both	Both	Both	Women	
20.	Training in solar engineering			Women	Women		
21.	Road construction[a]			Both	Both		

Legend: Women only; Men only; Both men and women; No intervention

[a] The Labour-Intensive Public Works (LIPW) component under the Ghana Social Opportunities Project (GSOP) provides short-term employment to extremely poor households to rehabilitate productive community assets such as feeder roads and dugouts (see Amoah & Mensah, 2023). In Zukpiri CREMA, both men and women were engaged and paid during the construction of the road leading to the CREMA entrance.
Source: Fieldwork data, July - August 2024

These interventions generate immediate short-term benefits, such as increased income (4.2.1) and financial relief through savings and loans (4.2.2), while also promoting long-term impacts, including community development projects (4.2.3).

4.2.1. Sources of income

Diverse livelihood interventions generate income for community members. First, honey and other bee products have provided income for many CREMA members. NGOs have connected beekeepers in some CREMA communities (e.g., Murugu and Nakong) to buyers for beeswax. For several reasons, income varies across communities, as beekeeping has been more successful in some (e.g., Murugu) than in others (e.g., Meguo). In most cases, beneficiaries of the beekeeping intervention receive two beehives and can earn an average of 69.74 USD (1000 GHS) annually if harvests are successful. However, exceptions exist. The individual cited below is an outlier, having expanded their beekeeping operation from the six beehives initially provided to about 90 beehives, which significantly exceeds the norm. This beneficiary highlighted the substantial benefits derived from the beekeeping project:

> "I can confidently say that the beehives are more profitable than even my farm work itself… I can boldly say that I have three children going to tertiary education and that I can pay their fees with the income from these beehives. Last year, after my sales, I bought this tricycle and this bike to support myself, so I use them to transport my hives from one place to another. They also help me with my farm activities." [Murugu male interviewee 3].

Those who participate in the establishment and management of tree nurseries are also paid directly for their work. In areas where dry-season gardens are established (Meguo, Siiru, Mognori), many women report earning income from selling vegetables and other staple crops.

Women acknowledge the role of NGOs and large-scale buyers in securing market access for non-timber forest products. These large-scale buyers pay market or above-market prices and commit to long-term relationships. Adopting a prefinancing model and paying a premium for the commodities have increased women's incomes (e.g., in Nakong). Before the CREMA interventions, women shea collectors struggled to find a reliable market for the shea they collected from the wild.

Furthermore, value-added processing of shea nuts in some communities has provided seasonal employment and income for shea-nut suppliers and shea butter processors working in the Murugu factory. The impact of the private-sector-led shea butter processing factory on women's income is highlighted by the company's representative:

> "So, recognizing that shea butter sold on a micro level, by the spoonful or by the kilo in a local market, is likely not going to be able to sustain your family as a woman. Because we buy such large volumes of shea butter, [we] allow the women who are part of the butter cooperative to bring more money home." [Private sector executive interviewee].

Further, many respondents noted that a new market has been created for some tree crops, such as baobab fruit, that previously had no economic value. Organizations such as ORGIIS Ghana purchase several bags of baobab fruit from women in SKGK CREMA each season. According to a male community member:

> "This baobab, nobody sells the fruit; they will just rot in the bush. But as this NGO came through the CREMA, a single baobab will not be left on the ground; they will pick all the good ones. So, the NGO will buy them and then export them." [Nakong male interviewee 9].

A female member of a neighboring CREMA community expressed

similar thoughts by noting:

> "Because of the conservation [through the CREMA], now we harvest the fruits of the baobab, sell them, and get money from it. Likewise, the tamarind and the shea." [Basisan female interviewee 1]

There is a disparity in shea-related interventions across the CREMAs. Although Zukpiri CREMA has organic certification, women in Siiru and Meguo still sell their shea nuts in the open market because there is no guaranteed corporate buyer, as in Nakong and Murugu, nor a shea-butter processing factory, as in Murugu. As a result, women in Zukpiri CREMA do not earn incomes comparable to those in Nakong or Murugu, where women receive a premium price and wages for working in the shea factory, respectively. However, work at the shea factory is seasonal, with a 1.5-months rotation cycle per women's group and the availability of shea nuts.

In Mognori, through the ecovillage, several thousand tourists have visited the community, generating income for both individuals and the community from tourist fees and gifts (Fig. 2[3]). Generally, project beneficiaries spend the money they earn on meeting daily and periodic basic family needs—paying children's school fees, buying school uniforms, meeting health needs, buying food ingredients, or investing in farm inputs.

4.2.2. Financial relief through savings and loans

VSLAs are present across the six communities but are not always associated with the CREMA program. In the SKGK CREMA, where they are widespread, the development of shea and baobab value chains, achieved by connecting nut and fruit collectors to the market, has led to increased savings within the VSLAs. This is also the case for women from Murugu, who deposit a portion of their shea factory wages into the *susu* group with the intention of taking home a substantial amount at the end of the year, when the money is disbursed. These VSLAs, predominantly composed of women, provide a low barrier to entry by allowing members to collectively set a minimum weekly contribution based on individual capacity. Members can borrow against their savings, often using the funds to pay school fees, access healthcare, or invest in farming activities. ORGIIS Ghana provides financial literacy and goal-setting training to these *susu* groups, encouraging reinvestment of income from sales and the productive use of their savings. To maximize impact, savings disbursements are timed to coincide with the farming season, ensuring the funds are allocated to farm inputs. Women in Nakong recounted the positive experiences they have had with the program:

> "Most women in the community belong to the *susu* group. If it's just you, you can only keep a little money, maybe 20 cedis, but by the end of the year, you'll be able to get 520 cedis plus interest. So, it's beneficial. You can take a loan from there. Do something to help yourself." [Nakong FGD Women]

4.2.3. Community development and other associated impacts

Many community development and social services have been brought to CREMA constituent communities through their participation in the CREMA program (see Appendix E in the Supplementary material for the list of community development projects). These initiatives are undertaken through two main pathways. First, because of the visibility the CREMA communities enjoy, numerous government agencies, NGOs, international organizations, and tourists visit and implement projects. These include the construction of a road in Zukpiri CREMA under a project jointly funded by the Government of Ghana and donors and the building of warehouses for commodity storage in SKGK and Murugu Mognori CREMAs through joint NGO, donor, and private-sector collaboration. In Mognori, foreign tourists have supported educational and medical programs, including paying school fees, awarding full scholarships to students, conducting medical outreaches, and donating clothing, bicycles, and school materials. UNDP Ghana has developed a long-term partnership with Zukpiri CREMA and has provided funding through the Global Environment Facility Small Grant Program (UNDP GEF-SGP) to support conservation and livelihood programs. Second, CREMA executives, through revenues generated by CREMA activities, fund programs to meet certain community needs. These include purchasing furniture for the Community Health Centre in Basisan, maintaining the ambulance in Nakong, and recruiting a paid volunteer to teach young children in Mognori. A community member in Meguo described the benefits of the community development projects as follows:

> "Through the CREMA, they gave us solar products; currently, as we talk, some people are still using the solar to even charge phones. We don't have lights here. Some students were supported with bicycles, especially those who trek from far distances to access the school here, which helped them come to school very early." [Meguo female interviewee 8].

Another impact is cost savings from using acquired skills to produce household goods. Women trained in soap-making produce soap in bulk for household use, reducing their expenses.

4.3. Distributional equity dimensions of livelihood interventions in CREMAs

The various interventions have had some impact on increasing individual and household incomes. However, these benefits are not equitably distributed across and within CREMAs, nor between men and women.

4.3.1. Uneven reach and impact of livelihood interventions across and within CREMAs

Despite the bold promises made by project implementers and the high expectations of local communities, the actual impact on livelihoods and well-being has been marginal, with only a few individuals in most CREMA communities benefiting from these interventions. Aside from shea-based interventions, which benefited many women, community development projects have been the primary means through which CREMA members have benefited indirectly. A respondent explained that:

> "A lot of programs come into the community that (…) should have made a great impact by now. Though they made an impact, it was not as expected, nor [was it] at a level that would alleviate the community from poverty." [Mognori female interviewee 9].

Recognizing that beneficiaries in CREMA communities have high expectations of improved livelihoods from the interventions, some state and non-state actors (including NGOs and donors) argue that these interventions are not intended to end poverty but to address basic needs. A key informant who shared this perspective emphasized that interventions are meant to create opportunities that will enable beneficiaries to turn things around for themselves. She said:

> "It's a false assumption to say that because we bring two beehives, this will change a person's life. It won't. We bring an opportunity and capacity-building. If the person commits to it, that person can make a difference in their own life and for their family. But no, the two hives themselves alone will not do anything." [Donor/NGO staff interviewee 7].

In only a few instances across the six communities have there been records of tangible, long-term improvements in livelihoods. In most

[3] The steep decline in tourist visits in 2020 was on account of the Covid lockdown.

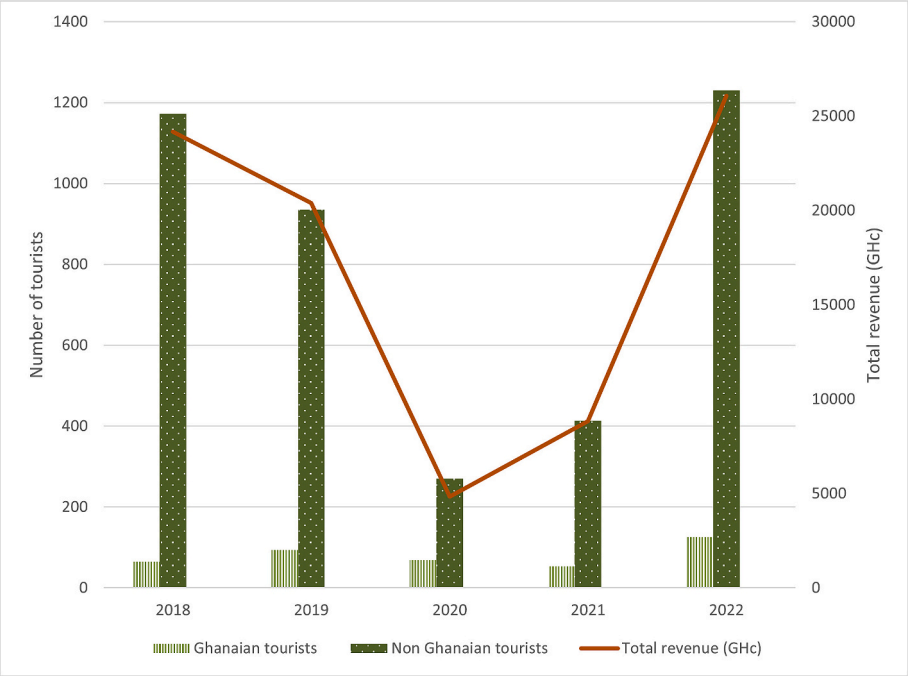


Fig. 2. Number of tourists and amount of revenue earned from 2018 to 2022 at Mognori Eco-village, Murugu-Mognori CREMA. *Source:* Authors' construct based on data provided by Mognori Eco-village Manager.

cases, the interventions have provided only a modest income supplement, helping individuals get by. The distribution of benefits is worse in CREMAs with numerous constituent communities, such as SKGK and Zukpiri CREMAs, which have 10 and 16 communities, respectively. The limited interventions are difficult to distribute across the large population, enabling a few individuals to benefit while others must wait several months or even years for subsequent disbursements of beehives or other donations. Consequently, progress in improving livelihoods is limited or unsustainable due to the inconsistency of interventions. As one respondent in Mognori puts it, "We rise, and we fall." Another respondent stressed that the interventions to date have not been successful in alleviating poverty.

> "To me, they [the interventions] have not yet gone an inch [in alleviating poverty]. There are many community members, and we have received only a few beehives. Even with that, we wait every 6 months before we can go there, probably to be lucky enough to harvest. How can that sustain a human being?" [Basisan male interviewee 3].

In contrast, Murugu Mognori CREMA, with only two communities, has reached more people through its interventions. Notably, the shea processing factory, which employs 80 women on a rolling basis, has had a positive impact on the community, with most respondents indicating that few households have been left out. A community member from Murugu explained that:

> "There has been an equitable distribution of CREMA projects here because I don't think there is a house in this village that has not benefited; either the man has beehives, and the wife is working at the shea [factory], or the wife is working at the shea [factory], and the man doesn't have beehives. Or they both have beehives, and the woman is still working at the factory, too." [Murugu male interviewee 3].

Cases of elite capture were observed, with those most active in the CREMA program, such as executives, patrol team members, and relatives of the local chief, often serving as beneficiaries of the interventions. For example, in Mognori Ecovillage, the late chief's house was selected as the first homestay for tourists, and the rooftop visited by tourists during village-guided walks is also within the same compound. In Siiru, the sole beneficiary sent to India for training in solar panel engineering, as part of a three-woman team from Zukpiri CREMA, was the late chief's wife. In SKGK CREMA, executives in the Nakong community were among the first to benefit from the initial batch of beehives supplied to the community. Respondents indicated that this was perceived as compensation for the voluntary work they perform as CREMA leaders, viewed as a token of "motivation" to continue their efforts. Similarly, interventions may become concentrated in the hands of early adopters within each community, often those with social capital and the ability to take risks, as seen with the introduction of beehives and ecotourism activities. In Mognori, the mother of the Eco-village manager sold shea butter to tourists, and her home was also used as a homestay. She noted that she combined both activities because most women were initially uninterested. After some women complained that she benefited more than others, she recommended including more women in such activities. Instances of favoritism or nepotism have led to the temporary or permanent abandonment of projects by community members. For example, in Mognori, the community abandoned the dry-season garden irrigation facility for over a year due to issues of favoritism and mismanagement.

4.3.2. *Gender disparities in livelihood interventions*

Gender plays a critical role in shaping both the implementation and impact of livelihood interventions. Most interventions have been specifically targeted toward women (see Table 3), resulting in greater benefits for women in the CREMA communities. Conversely, leadership positions within CREMAs, at both the community and CREMA levels, remain male-dominated, with women comprising only 15–30% of leadership roles across the three CREMAs studied. Consequently, men wield significant decision-making power in key aspects of livelihood interventions, including transportation, warehouse and factory operations, and ecotourism enterprises. In contrast, holding a CREMA leadership position does not necessarily grant women the same level of influence. A female member of the CREMA executive highlighted this disparity:

"As for here, women have no power to always make decisions like that." [Basisan female interviewee 6].

Meanwhile, NGO efforts to mainstream gender into interventions to empower women and to align interventions with women-based livelihood activities, such as the collection and processing of shea nuts, have prompted increased focus on women's livelihood development. Even in livelihoods where men might have a comparative advantage, such as beekeeping (as women often fear being stung), women continue to participate with the support of men, including their husbands or adult male children. A male participant emphasized this:

"The women benefit more than the men because they are involved in the shea activities, where they sell to the people, and they also work in the factory, where they produce again. So, at the end of the two weeks, sometimes each woman could get like 400 Ghana cedis after sharing the proceeds. And some of them are also still involved in the beehives as well." [Murugu male interviewee 1].

The interventions have contributed to the economic empowerment of women in the CREMAs by increasing their income-generating capacity and enabling them to meet some household needs and care for their children, rather than relying entirely on their husbands.

"It [shea trade] is actually improving us. See this building; a woman built it. I'm building [too] because of this shea. Because I don't sell my shea to the market again, neither will they collect lorry fare. So that money I used to go and do *susu*, and I get about a thousand cedis. I go and use it to buy cement, come and dump it, mold blocks, and use [the remaining] for my children's school fees, hospital, and for my farming." [Nakong female interviewee 10].

Men recognize the positive impact of the interventions on women. During one of the men-only focus group discussions, a participant noted:

"If you empower a woman, you have empowered a whole family. But if you empower a man, you have just empowered that individual. In the garden, when it was working, the women used to sell the vegetables and had some money. The men even used to borrow money from the women. So if you borrow your wife's money and don't pay, can she send you away?" [Meguo male interviewee 5].

However, prevailing cultural roles and norms, especially in northern Ghana, compel women to engage in these livelihood interventions while also meeting their household obligations and demands. As a female interviewee bitterly explained:

"Most of the household responsibilities are always done by women. The men don't really have that responsibility because they just bring foodstuffs, and then you cook; it's not their business again. So I think most women are up and doing; they want to get involved in everything that will earn them money so that they can take care of their families." [Murugu female interviewee 9].

A male member of the CREMA community also supported this:

"I feel like we, the men too, need to be up and doing so that once the women make a profit, we will also make a profit, which will uplift us very high. But most of us depend on women to make the profit, and take care of everything, not that way." [Murugu male interviewee 3].

Many female respondents advocated for women-centered interventions because they have the greatest impact on the household as a whole. A female executive member in Murugu is a strong advocate:

"For a community like ours, the best alternative is to center everything on the women because the man can decide to have two or three wives, and the only thing he does is provide foodstuffs to be cooked. Anything else, like family healthcare, is not really a major concern to him. So once the women are empowered, it will change the household more than the men." [Murugu female interviewee 5].

While most male respondents agree that women should benefit from existing livelihood interventions, they expressed hope that men will not be completely excluded as beneficiaries of future interventions.

5. Discussion

5.1. Livelihood interventions in context

Our study identified diverse livelihood interventions implemented across three CREMAs in northern Ghana, generating impacts at both the individual and community levels. These interventions largely mirror those documented across CREMA communities (Adeyanju et al., 2021; Agyare et al., 2015a, 2015b, 2024) and, more broadly, in CBNRM programs across Africa (Charnley, 2023; Wicander and Coad, 2018; Roe et al., 2015). Consistent with prior research, most interventions focused on enhancing or diversifying existing livelihood activities rather than introducing entirely new ones—a strategy widely regarded as less risky and more culturally acceptable (Torell et al., 2010; Sene-Harper et al., 2019; Wright et al., 2016). Interventions targeting existing livelihoods build on established skills and knowledge, leading to faster adoption, reduced risk, and greater community ownership (Sene-Harper et al., 2019; Torell et al., 2010). This reinforces the argument that securing local acceptance is critical to the success of interventions (Roe et al., 2015; STAP, 2024; Wright et al., 2016).

However, our findings also underscore persistent tensions between community expectations and actual outcomes. While CREMA implementers often promise improved livelihoods to encourage participation, conservation objectives frequently take precedence, and livelihood interventions are introduced later, often irregularly, and with limited scope (Bempah et al., 2019; Mawutor and Hajjar, 2024). This pattern contributes to the expectation-performance gap observed in previous studies (Agyare et al., 2024; Ahmed and Gasparatos, 2020; Bempah et al., 2019). In some cases, communities join CREMAs for non-economic reasons, such as wildlife protection in the Kunlog (Jelinkon) CREMA, yet economic outcomes are expected as a by-product of their participation (Robinson and Sasu, 2013). These findings highlight the complexity of defining community objectives, which span environmental, economic, social, and cultural domains (Adeyanju et al., 2021; Murphree, 2009; Berkes, 2013).

Governance challenges further complicate outcomes. Elite capture and nepotism, widely reported in CREMAs (Baruah, 2017; Bempah et al., 2019) and other CBNRM programs (Jusrut, 2022; Persha and Andersson, 2014), were evident in our case study sites, raising concerns about equity and fairness. Moreover, the sustainability of interventions remains precarious because of heavy reliance on external funding, which limits long-term benefits and risks beneficiaries reverting to previous poverty levels once a specific project cycle end (Adeyanju et al., 2021; Mawutor and Hajjar, 2024). Our findings echo broader critiques of the persistent risk of elite capture in incentive-based conservation and the limitations of donor-dependent interventions.

5.2. Linking gender and contexts with distributional equity in livelihood interventions

Our findings indicate that most livelihood interventions across the CREMAs targeted women exclusively or in combination with men. This aligns with global trends in rural development projects, where conservation development organizations increasingly promote women's inclusion (Westerman, 2021). Advocates argue that integrating gender equality and women's empowerment into poverty-reduction strategies is essential, given women's disproportionate vulnerability to poverty (Awumbila, 2006; Reed et al., 2024; Westerman, 2021). In our case,

interventions that promote market access and value addition for shea and other economic trees (resources culturally associated with women whose harvesting of nuts and fruits align with conservation goals) provide a valuable source of cash income to rural women who otherwise have very few income options. This aligns with findings from Ghana (Gilli et al., 2020; Sheppard et al., 2010), Burkina Faso (Ingram et al., 2015; Pouliot, 2012), and Mali (Naughton et al., 2017).

Beyond gender inclusion, distributional equity norms shape how benefits are allocated. These norms include the proportionate distribution of benefits among participants (egalitarian approach), distribution in proportion to contributions and costs (merit-based approach), and distribution that addresses the poor's welfare needs (pro-poor approach) (Friedman et al., 2022). The emphasis on women in our study sites reflects a pro-poor approach. However, equity is not only about net-positive outcomes but also about fairness, which varies across contexts (Friedman et al., 2022; Martin et al., 2019). For instance, Martin et al.'s multi-country research revealed that people's preferences for distributional norms vary based on whether benefits are monetary or non-monetary. Participants were less inclined to favor 'pro-poor' or 'equality' principles when it came to distributing forest revenue (Martin et al., 2019). Similarly, Abebe et al. (2020) report that age, gender, and social roles shape perceptions of fairness in Ethiopia. Our findings echo these patterns. Women in CREMAs prefer that livelihood interventions target them, noting that benefits are allocated to meet household needs rather than solely their personal needs. While men supported women's inclusion in these interventions, they also expressed concerns about men's exclusion.

These dynamics underscore a critical tension: while equity is assumed to enhance the success of interventions, excluding influential individuals or groups can undermine project outcomes (Klein et al., 2015). Hence, pro-poor distribution norms that favor women over men in power may prove counterproductive to overall project success. Moreover, treating women as a homogeneous group without considering intersectional factors, such as wealth, age, religion, and social status, risks reinforcing inequities (Lau, 2020).

Finally, our study highlights the challenge of achieving fair distribution in multi-village CREMAs. Because interventions are insufficient to meet demand in the first place, distributing them across multiple communities benefits only a few individuals, leaving others waiting for future opportunities that may never materialize. Similar patterns have been observed in CBNRM projects in southern Africa, where single-village initiatives yield higher benefits than multi-village contexts (Child and Barnes, 2010; Suich, 2013). This reinforces the importance of considering contextual factors in designing equitable interventions (Abebe et al., 2020; Friedman et al., 2022; Martin et al., 2019).

5.3. Balancing conservation and human livelihoods in CREMAs

Here, we re-examine the synergies and trade-offs of integrating conservation and development. In our CREMA case study, we observed a strategy that Salafsky (2011) describes as "explicitly linking the project's conservation ends to broader development ends." In these CREMAs, conservation goals have been leveraged to achieve development outcomes through livelihood interventions, suggesting a potential synergy between these domains. To illustrate this, we present two key examples of synergies from our findings.

First, introducing modern beekeeping practices has significantly reduced the need for traditional wild honey harvesting, which often involves felling trees and using fire to smoke out bees—practices that contribute to deforestation and wildfires. By shifting to sustainable beekeeping, trees are preserved, and biodiversity is supported through the protection of bee populations. At the same time, local beekeepers benefit economically from the sale of honey and beeswax, creating a direct link between conservation and improved livelihoods. Second, the CREMA program has also helped protect shea trees, which are often cut down for charcoal production, thereby conserving a key economic resource crucial to women's livelihoods. Conservation efforts have been complemented by reforestation initiatives to plant new shea trees, ensuring long-term sustainability. Moreover, women engaged in tree nurseries and the establishment of plantations receive financial compensation, further reinforcing the development benefits of conservation.

Many conservation actors in Ghana consider CREMAs a viable platform for reconciling biodiversity conservation with community development (Noé Ghana, n.d.). The synergies discussed previously are encouraging examples of this reconciliation in practice, offering models that merit strengthening and replication across CREMAs.

5.4. Study limitations

We acknowledge a few limitations that may have influenced the findings. First, because government agencies, donors, and non-governmental organizations implement multiple conservation-development interventions across the region simultaneously, participants may have mistakenly attributed the impacts of unrelated initiatives to the CREMA program. Second, participants may have had difficulty accurately recalling specific details, such as the year interventions began, their duration, and the amount of income generated. Finally, despite clearly informing participants that the research was conducted as part of an academic project, there remains the possibility of response bias. For instance, CREMA executives may have overstated the program's positive impacts to present CREMA in a favorable light, while other community members may have understated the benefits to emphasize the need for additional livelihood support.

6. Conclusion

The CREMA concept emerged as a key natural resource management program, recognized both within and beyond Ghana. Among its main objectives, improving livelihoods through the sustainable use of natural resources and the deployment of supplemental income-generating interventions have been pursued.

In general, most livelihood interventions have modestly improved people's incomes, though only seasonally. Hence, there is a need to explore interventions that can offer regular, long-term income or employment for CREMA members. Similarly, some interventions have been more successful and impactful than others. In particular, interventions that modernize traditional livelihood activities and practices (e.g., the use of beehives and organic shea trade), provide market access (e.g., a guaranteed private-sector buyer, such as Savannah Fruits Company), and add value at the local level (e.g., the shea butter factory in Murugu) have had the most impact and success across all CREMA communities, especially for women. Nonetheless, the spread of benefits remains limited due to elite capture, poor project execution, limited quantity, and the irregular frequency of the interventions, among other challenges. In addition to direct benefits related to improved income, some community development projects have been carried out as a by-product of the CREMA program, which has been the main pathway through which benefits reach most people within the CREMAs.

With most interventions adopting a gender-targeted, pro-poor distribution norm, women's inclusion and access to cash income improved, but this progress raised concerns about potential risks of men's exclusion and other intersectional inequities. The context in which interventions are implemented also influences fair distribution, especially when multiple villages form a single CREMA.

To advance CREMA's livelihood improvement objectives, we propose the following recommendations for project designers and implementers across government, donor, and non-governmental organizations. First, implement robust community engagement processes to identify interventions that align with local preferences. Such engagement fosters ownership and strengthens long-term commitment to project sustainability. Second, prioritize market integration and value addition for

economic commodities within CREMA landscapes. Equitable market access is essential for enhancing household incomes and income security. Third, while diversification into green value chains is crucial, particularly in a changing climate, agriculture remains the primary source of income for most households. Therefore, livelihood interventions should complement diversification with targeted agricultural support and investment to enhance farm profitability and climate resilience. Fourth, the government and its development partners should increase financial commitments to ensure interventions are consistent, scalable, and capable of delivering measurable impact across a significant portion of each CREMA community. Securing equitable partnerships with the private sector could help inject new or untapped funding sources, potentially helping address donor dependency in the long term. Finally, capacity-building initiatives must extend beyond skills training to include guaranteed start-up capital and ongoing technical assistance. This integrated approach will enable beneficiaries to establish viable, sustainable income-generating activities.

Ethics approval

This study was approved by the UBC Behavioural Research Ethics Board (H23-02989).

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Declaration of generative AI and AI-assisted technologies use

During the preparation of this work, the authors used Microsoft Copilot and Grammarly to improve grammar and phrasing. After using these tools, the authors reviewed and edited the content before inclusion in the manuscript. We take full responsibility for the content of the published article.

Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary material to this article can be found online at https://doi.org/10.1016/j.crsust.2026.100337.

Data availability

Data will be made available on request.

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